

EXPERIMENT - 2

Design and configure the Switch network using cisco packet tracer

AIM :

To design and configure a basic Local Area Network (LAN) using a Switch-PT and five PCs in Cisco Packet Tracer, and to assign Class A IP addresses so that all devices are on the same subnet and can communicate with each other.

PROCEDURE :

1. Physical Setup

1. Open Cisco Packet Tracer.
2. From the Network Devices section, drag and drop a Switch-PT into the workspace.
3. From the **End Devices** section, place **five PCs** and label them as **PC0, PC1, PC2, PC3, and PC4**.
4. Select Copper Straight-Through cable.
5. Connect:
 - PC0 → Switch (FastEthernet0)
 - PC1 → Switch (FastEthernet1)
 - PC2 → Switch (FastEthernet2)
 - PC3 → Switch (FastEthernet3)
 - PC4 → Switch (FastEthernet4)

2. Hardware Modification

1. If the switch does not have enough ports:

- Click on the **Switch-PT**.
 - Go to the **Physical** tab.
 - Turn **Power Off**.
 - Insert **PT-SWITCH-NM-1CFE** module into an empty slot.
 - Turn **Power On**.
2. Use the newly added ports to complete all PC connections.

3. Logical Addressing

Configure IP addresses for each PC:

PC	IP Address	Subnet Mask
PC0	10.1.1.1	255.0.0.0
PC1	10.1.1.2	255.0.0.0
PC2	10.1.1.3	255.0.0.0
PC3	10.1.1.4	255.0.0.0
PC4	10.1.1.5	255.0.0.0

Steps for each PC:

1. Click on the PC.
2. Go to **Desktop → IP Configuration**.
3. Enter the assigned **IP Address**.
4. Set **Subnet Mask** to **255.0.0.0**.
5. Leave **Default Gateway** empty (not required for same-LAN communication).

4. Verification

Open **Command Prompt** on any PC.

Use the **ping** command to test connectivity:

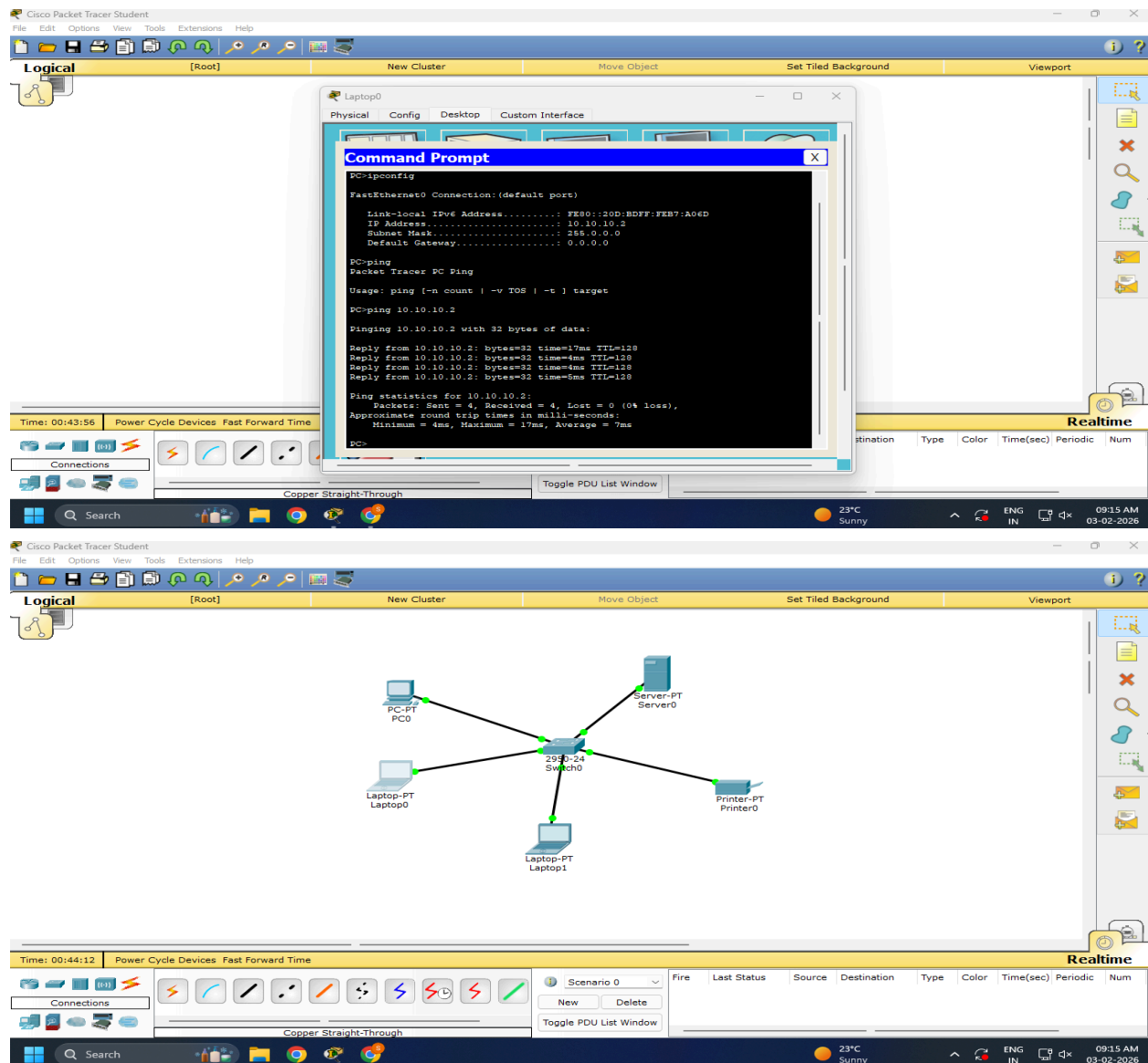
```
ping 10.1.1.2
```

```
ping 10.1.1.3
```

```
ping 10.1.1.4
```

```
ping 10.1.1.5
```

OUTPUT :



RESULT :

A LAN was successfully created using a Switch-PT and five PCs. All systems were assigned Class A IP addresses with the subnet mask 255.0.0.0, placing them in the same subnet. Communication between all PCs was verified successfully using the ping command.

